

2D RPG VIDEO GAME WITH THE UNITY ENGINE

Prepared By

Matthew Wempe

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Matthew Wempe

Dr. Sean Hayes

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The project is a top down Two-Dimensional (2D) Role-playing (RP) dungeon crawler game. The player takes control of a mage who traverses through dungeon levels while encountering challenging enemies and bosses. Along their journey, players can expect to collect gold and gain experience which can be used to level up their health and skills. For players to advance to the next dungeon level, they will have to complete certain objectives. These objectives vary from defeating a certain number of enemies to solving puzzles.

Software that was needed for this project was the Unity Engine that the framework that the game is built upon. The Unity Engine utilizes C# scripts which the project was coded in. A crash course of Udemy taught me how to traverse and use the Unity Engine for game design.

Role Playing Games have started to lose their unique styles and depth of story that they previously had. For example, the Fallout franchise suffered a backlash for promoting Fallout 4 as an RPG, but replaced Role Playing elements with more resource management and a shooter playstyle. Also, the quests that lacked story. Ubisoft has also started to shift its RPG franchises, Dragon Age and Mass Effect, into fast RPGs with repetitive content that have no sense of reward. People are ditching the role-playing genre altogether because the immersion they once had is no longer present. In recent years, 2D games have started to see a rise in popularity due to their retro look and feel. To draw people back to the RPG genre I have come up with a solution: a fast paced 2D RPG game with quests that provide depth and a sense of accomplishment. Players will be able to customize their character along with changing and upgrading their skills at will. This game will bring back the wonder that was first captured by the early games of the Fallout, Dragon Age, and Mass Effect franchises.

Ideally, I would be able to develop a game with my current state of knowledge, however, there is knowledge that I lack when it comes to video game development. This will be a hurdle I need to overcome in order to be able to use game engines like Unity. Overcoming this problem, I will need to take extra classes outside of my normal schedule to increase my comprehension surrounding video game development. Topics that will assist me in this pursuit include: animation, C# scripting, collision, and physics. These topics can also be applied to other projects that I might face further down in my career.

When games are developed, it takes multiple teams that specializes in different areas, such as art design, dungeon layout, UI/UX design, and scripts. Due to my organization progress and scrum methodology, I managed to keep myself accountable and develop a functioning game. Software design is not just coding, it includes the design itself. This project provided an opportunity not offered in my coursework to learn about game design and development, the largest and most profitable fields the tech industry.

My future research efforts will be to continue to add to the game over the course of the next couple of years. The following features are what I wish to implement:

-Dynamic artwork

This will help me better visualize my game and give smoother transitions. This is a challenge for me because I am not an artist and would require hiring an artist to create the pixel art for the game.

-Animations for Combat

This will help polish the game and make it look and feel less clunky. The drawback is that I currently lack artwork to do this step.

- Dialogue System

Implementing dialogue and populating the world with NPCs to give the game more depth. A challenge that must be overcome for this task is writing a dialogue and coming up with a story along with choices for the player to make.

Deliverables: The first and only deliverable for 498 will be submitted on August 20th. Almost all the coding for my game has been completed over the summer. The deliverables for 499 will be spread out every two weeks with general bug fixes and tweaking until the final presentation of the game at the end of the semester. The deliverables will be as followed:

May 12th - Complete Udemy course.

May 14th - Began on spell framework.

May 17th - Completed spell framework and finished the first spell.

May 18th - Created the Spell UI.

May 19th - Began connecting the spell UI to buttons and coding backend.

May 21st - Spell UI fully implemented.

May 22nd - Began work on "Muck" Enemy.

May 25th - Finished working on "Muck" Enemy.

May 26th - Started Upgrade Spell and Equip Spell design.

May 28th - Finished Upgrade Spell.

May 29th - Finished Equip Spell and began working on Spell Hotbar.

May 30th - Finished Spell Hotbar.

June 2nd - Began working on Player Spellbook.

June 5th - Finished Player Spellbook and removed features from Udemy Course.

June 7th - Started work on new combat system.

June 10th - Finished combat system.

June 12th – Created Damage Over Time Spell.

June 16th - Added in dozens of sprites and created a new level.

June 21st - Begin work on new enemy Mage.

June 30th - Finished Mage.

July 5th - Redesigned HUD.

July 6th - Redesigned the saving state of the game.

July 8th - Tweaked Combat.

July 13th - Enhanced damage system.

July 14th - Started the AOE spell.

July 15th - Finished the AOE spell.

July 22nd - Redesigned

July 23rd - Redesigned Menu.

July 25th - Created Health Potion.

July 28th - Started work on “Gosu” Enemy.

July 31st - Finished work on “Gosu” Enemy and created “GosuSmall” Enemy.

August 1st - Created Doors and tweaked the Mage.

August 3rd - Started work on Objective Manager.

August 7th - Finished Objective Manager and created objectives.

August 10th - Started Tutorial Manager. Started and finished Tutorial Class.

August 16th - Finished Tutorial Manager.

August 17th - Created the first Tutorial Level.

August 20th - Created the second Tutorial Level.

September 3rd - Create the third level for the game.

September 12th - Tester one runs through course

September 13th - Feedback from tester one is implemented

September 14th - Tester two runs through course

September 15th - Feedback from tester two is implemented

September 17th - Tester three runs through course

September 18th - Feedback from tester three is implemented

September 19th - Tester four runs through course

September 21st - Feedback from tester four is implemented

September 22nd - Tester five runs through course

September 24th - Feedback from tester five is implemented

September 25th – Finish final usability study and evaluate results.

October 1st - Polish up levels with content.

October 15th - Bug fixes and tweaking.

October 29th - Bug fixes and tweaking.

November 8th – Defense run-through.

November 12th – Defense

Project Requirements

The Two-Dimensional Role-Playing Game required the following to create:

1. Computer
2. Game Engine – Unity Engine
3. Programming Language – C#
4. Artwork – Royalty-Free artwork

The Unity Engine was chosen because it is becoming one of the most popular game engines for indie games along with being very user friendly for beginners. Unity offers a plethora of helpful tutorials and documentation to teach aspiring game developers. C# was chosen for the following reasons to code my game in: C# is becoming the video game industry's standard language for coding and Unity allows default C# support. Along with C# being a strong choice, almost every video game job application asks for knowledge of this language.

With having a very small budget for my project, I cut budget costs by not hiring an artist to create custom artwork and sprites for my game. Instead, I found royalty-free artwork online that I could use.

One feature that I wanted to implement in my game was damage over time. I started this implementation with a spell game object. The goal was that the damage over time spell would collide with an enemy, the spell would do a certain amount of damage every other second. Unfortunately, this implementation failed because it would distribute all the damage at once.

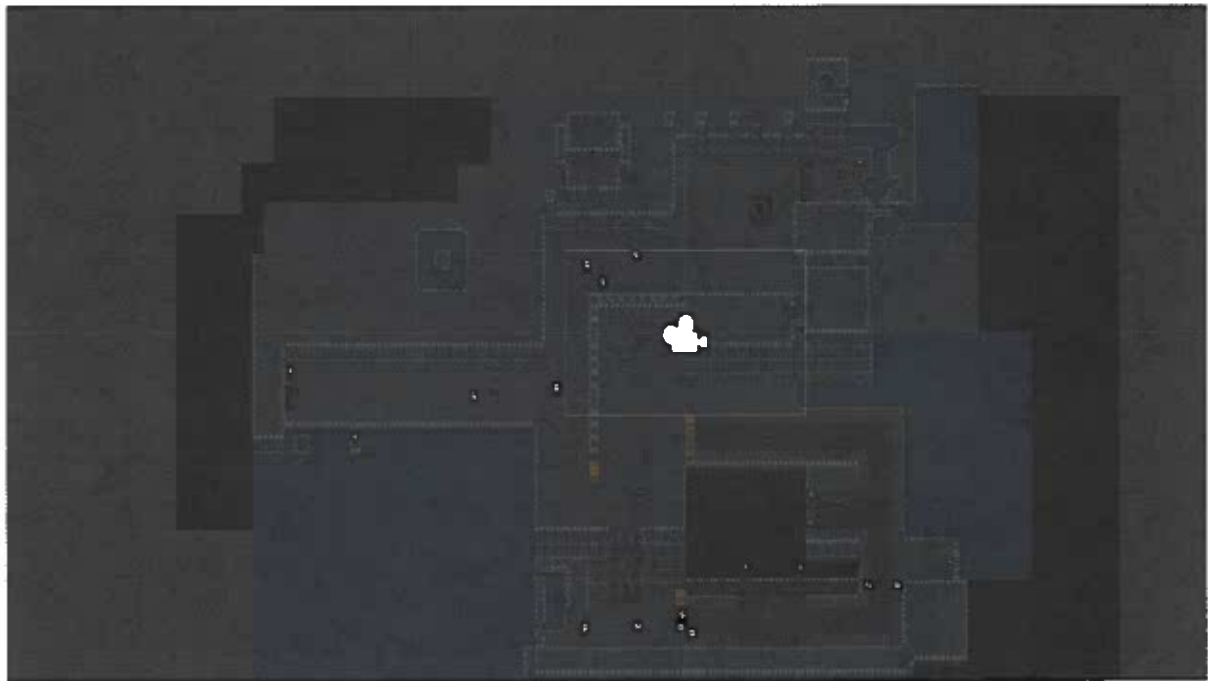
At this point in time, I had not taken Operating Systems and had not learned how threads operate. After doing some research I came across an online post about someone wanting to run a countdown but kept failing because the game wouldn't allow any other scripts to be ran at the same time. Someone replied that all the person had to do was implement a coroutine. This allows the game to be running multiple scripts simultaneously. After reading the post, I quickly implemented how I wanted my coroutine to run and I was successful in creating my damage over time spell.

```
48 IEnumerator RecieveDamageOver(DamageOverTime dmg)
49 {
50     onDot = true;
51     int currentCount = 0;
52     while (currentCount < dmg.damageCount)
53     {
54         hitPoint -= dmg.damageAmount;
55         GameManager.Instance.ShowText(dmg.damageAmount.ToString(), 25, Color.red, transform.position, Vector3.zero, 0.5f);
56
57         if (hitPoint <= 0)
58         {
59             onDot = false;
60             hitPoint = 0;
61             Death();
62         }
63         yield return new WaitForSeconds(dmg.time);
64         currentCount++;
65     }
66     onDot = false;
67 }
68
69 protected void StartCorou(DamageOverTime dmg)
70 {
71     StartCoroutine(RecieveDamageOver(dmg));
72 }
73 }
```

Screenshots from Game







Test Plan Identifier:**Wempe 2D RPG Feedback Tests****Introduction:**

The testing entailed running five different people through different revisions of the game. Every tester provided their feedback on the game and how they felt about their experience. This feedback was implemented into the 2D RPG game before another tester was run through the process. The main objective of this process was to refine the game as much as possible. It helped shed light on what is appealing to users and what features of the game need more refinement. Originally, I worried that the constraints for this testing would include: Testers might not provide valuable data that can be implemented and not all feedback may be implemented due to time constraints.

References:

Original Questionnaire

Completed Questionnaires done by testers

Project Plan

Test Items:

Wempe 2D RPG v0.1

Features to be tested:

Movement

Spells

Tutorial

Enemies

Portals

UI

Features Not to Be Tested:

The game was tested so no feature was exempt from testing.

Item Pass/Fail Criteria:

Pass/Fail was tested on whether feedback showed that testers complained about the same feature. That showed that work is not being done to improve the feedback from the previous tester. Pass criteria will be based on whether feedback was implemented and that future users gave feedback concerning other areas of the game.

Suspension Criteria and Resumption Requirements:

Testing was suspended after every tester had run through the game and filled out their questionnaire. Feedback was reviewed and then implemented into the game.

When testing resumed, the new tester was given the questionnaire and a copy of the game. They were required to fill out the questionnaire.

Test Deliverables:

- Test Plan (this document itself)
- Questionnaire
- Completed Questionnaires
- Test Script
- Test Report
- Defect/Enhancement Log

Test Environment:

The tester had to have the following: a computer and a copy of the game in order to test the content.

Estimate:

The testing took thirty minutes for each user to run through. Afterwards the questionnaire was filled out within roughly five to ten minutes. Implemented feedback could take up to two hours to implement and test before being finalized. Total time took around thirteen to fourteen hours. This time was spread out through multiple weeks.

Schedule:

The test schedule:

September 12th - Tester one runs through course

September 13th - Feedback from tester one is implemented

September 14th - Tester two runs through course

September 15th - Feedback from tester two is implemented

September 17th - Tester three runs through course

September 18th - Feedback from tester three is implemented

September 19th - Tester four runs through course

September 21st - Feedback from tester four is implemented

September 22nd - Tester five runs through course

September 24th - Feedback from tester five is implemented

Staffing and Training Needs:

Only staffing needed was done by the creator, Matthew Wempe.

Skills required was knowledge of C# and Unity Engine.

Responsibilities:

Matt Wempe - Send questionnaires and game to volunteer testers. Receive feedback from finished questionnaires and implement changes into game.

Risks:

Only risk that could be seen is that of a tester backing out of wanting to participate. If this occurs, there were backup testers to fill their place.

Approvals:

Only approval necessary was that of the testers, which was taken before the testing began.

Test Script

The process that I ran through each participant that took part in my testing remained almost consistent throughout the entire testing process. I felt the most valuable feedback I could get were from people that I knew who played video games regularly. I contacted my closest colleagues to see if they would be interested in testing out my game and giving me feedback. I consulted with each one of them and scheduled them throughout September 12th through September 22nd.

On the day of the testing I sent each tester a version of the game for them to download onto their computers to test the game. I also attached the questionnaire for them to fill out immediately after playing the game. We entered into a discord call, so I could be standing by to answer any questions about the mechanics of the game if they had any. After they finished the game, the tester filled out the questionnaire and sent it back to me via email.

After reading through their questionnaire, I implemented changes based on their feedback and fixed any bugs that they had encountered during their test. Once this step was complete, I would start over the same process and send the revised game to the next tester. I would then implement their feedback until I had gone through all five testers.

Test Script Broken Down

1. Obtain testers consent.
2. Schedule testing time.
3. Day of test the game and questionnaire were emailed to tester.
4. Setup communications to answer questions for tester.
5. Tester completed their playthrough of game.
6. Tester filled out questionnaire.
7. Questionnaire was sent back for further review.
8. Feedback from questionnaire was implemented into game.
9. Started step 3 with the next tester.

Test Report

This document is a brief summary of the questionnaires that were filled out from the testers who ran through my game and the feedback that was implemented from the questionnaires.

The first tester commented on enjoying the combat system but felt the game's tutorial took too long to finish. Taking their feedback into consideration I implemented a new feature that allows manual skipping of the tutorial. This vastly improves the gameplay of the game by letting veteran players or people who have are familiar with the game, be able to skip through the information that they already know. This allows faster access to the content that they players want to enjoy.

After implementing the tutorial skip, the second tester was given the improved version of the game. The second tester utilized the new skip button for the tutorial system and did not mention anything about the tutorial being too slow. They enjoyed the wide variety of enemies that the game offered but found the player's spell book to be too encumbered. They did not like all the choices of spells that they could not equip. For background, I had initialized entire spell tress, but never implemented any of the spells to work. To fix this, I removed most of the spells in the spell trees to only have the ones the players can use.

The third tester's feedback did not include anything regarding the spell tree. Their main complaint was with the artwork that the game had. This was the hardest feedback to implement because of two things: I am not an artist so hand drawing the art for the game is beyond my scope. Secondly, if I were to obtain better artwork, I would need it commissioned by an artist. The artwork that I have found so far has been royalty free, however finding art to match up with what I currently have is very difficult to find. The changes that I did do was add in better artwork for the spells and the health potion.

Once the updated artwork was implemented the fourth tester had their run through. The fourth found a very interesting bug with the "small Enemy" that occurred when the small enemy came into melee range of the player. The enemy would instantly kill the player due to the enemy not having a cooldown on their attack after every swing. This bug was fixed by adding a cooldown for the enemy's attack.

The final tester ran through the entire game without problems with the tutorial manager, spell book, or getting instantly killed by an enemy. The major complaint that the fifth tester found with the game was spell animations. So better spell animations come with more artwork, and since I lack artwork this part was very difficult. What I did do to improve the spell animations was slow down the spell speed to make the spells look more fluid. It was not a perfect fix, but an improvement to what it was.

The testing provided great feedback to the game that I have such a deep passion for. To be able to share what I have created with others has been a wonderful experience. This testing helped me flush out the strengths and weaknesses of my project and that it can only get better with future testing.

Age: 22

1. How often do you play games?

Everyday

2. What is your favorite genre of games?

Battle Royale games

Thoughts

1. What did you enjoy about the game?

I really liked the combat. I like that you shoot where your mouse it at. Artwork is very cool too.

2. What feature(s) did you find tedious or confusing? Why?

I didn't like having to wait through the tutorial. I'm an experienced gamer so I know how most RPG systems work.

3. If you designed the game, what aspect would you have changed?

Tutorial

a. How would you have changed it?

I would have made it so you could skip it.

Age: 22

1. How often do you play games?

Everyday

2. What is your favorite genre of games?

Third-Person shooters

Thoughts

1. What did you enjoy about the game?

I had a lot of fun. It was really cool to see all of the enemies and to think that there is more to come is pretty cool.

2. What feature(s) did you find tedious or confusing? Why?

Probably the extra spells that I couldn't use. Just because they are shown and I can't equip them.

3. If you designed the game, what aspect would you have changed?

Spells

a. How would you have changed it?

Remove the spells

Age: 22

1. How often do you play games?

Everyday

2. What is your favorite genre of games?

MMO RPG

Thoughts

1. What did you enjoy about the game?

The mechanics are pretty cool and the enemies are very interesting.

2. What feature(s) did you find tedious or confusing? Why?

Nothing was tedious

3. If you designed the game, what aspect would you have changed?

The artwork

a. How would you have changed it?

I would have drawn much better pictures for you to use.

Age: 22

1. How often do you play games?

Everyday

2. What is your favorite genre of games?

FPS

Thoughts

1. What did you enjoy about the game?

I liked the artwork and the spell tree.

2. What feature(s) did you find tedious or confusing? Why?

Enemies can sometimes one shot you.

3. If you designed the game, what aspect would you have changed?

Make it so enemies can't one shot you.

a. How would you have changed it?

I don't know how since I don't make video games. But I would find a way to change that.

Age: 22

1. How often do you play games?

Everyday

2. What is your favorite genre of games?

FPS, Battle Royale

Thoughts

1. What did you enjoy about the game?

The game felt pretty fluid from the first time I tried it.

2. What feature(s) did you find tedious or confusing? Why?

Not much. The spells don't look that good.

3. If you designed the game, what aspect would you have changed?

Made the spells have special effects when you shot them

a. How would you have changed it?

Hired an artist or animator for the spells to make them flashy.

Defect/Enhancement Logs

09/13/18 – Implemented a tutorial skip. This allows for the user to not have to read through the text that is displayed.

09/15/18 – Fixed bug that was instantly killing player when attacked by a small enemy.

09/18/18 – Implemented in more artwork.

09/21/18 – Made extra spells not available to user. Will be implemented at a later date, so not fully removed.

09/24/18 – Smoothed animations of the user interface.